Decision Making Using Marketing Analytics

MKT 566 · Fall 2026

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USC Marshall School of Business

Introductions

A little about me

- **11th year at USC** (office HOH 332)
- During **2023–25**, worked full-time at **Amazon**, first in Ad Measurement, then in Climate Pledge Friendly
  - Worked closely with data scientists, research scientists, sales teams, and product managers
- Currently part-time at **Airbnb** as *Housing Scholar*
- I run the Real-Estate Analytics Lab (REAL) and its free Substack with Marco Giacoletti

> Personal website: https://dadepro.github.io/.

A little about me

Research on **online marketplaces and policy**:

- Trust and reputation
- Platform manipulation
- Short-term rentals (Airbnb) and real estate
- Advertising
- Sports betting

My website has links to all my papers. They are also on SSRN.

A little about you

- What is your background?
- What do you expect from this class?
- Have you worked in marketing yet?

The Course

Who to contact

	Instructor	Teaching assistant
Name	Davide Proserpio	Raghav Sarmukaddam
Office	HOH 332	HOH 332
Email	proserpi@marshall.usc.edu	sarmukad@usc.edu
Hours	Tue 2–4 pm	Wed 2:30–4:30 pm

Both by appointment as well.

Organization

Schedule. 11:00 am – 12:20 pm, Tuesday and Thursday, in **JKP 202**

Class website. https://github.com/dadepro/mkt566

Syllabus. mkt566-syllabus-proserpio.pdf

> Most lectures involve examples and exercises. Some classes are given over entirely to exercises, discussion, or guest speakers.
>
> I am also planning to have a few guest speakers, very likely from tech companies.

This class is hands-on. Bring your laptop and be prepared to analyze data, write code, and present results.

Readings

Slides, plus open-source reading links provided for each topic.

Open source

- My course on data storytelling
- Data Visualization: A Practical Introduction
- R for Data Science
- R Markdown
- R for Marketing Students

Not open source (slides, exercises, and code are)

- R for Marketing Analytics
- Python for Marketing Analytics

Analyzing data and coding

- I will use **R and Python**
  - R primarily for data visualization and statistics
  - Python primarily for machine learning
- For assignments you may use either, but following what I do is easier, especially if you are new to coding

IDE suggestions: VS Code (pretty much any programming language, integrates with Codex and Claude), RStudio (R only), Cursor.

Setup

Before the next class, install:

1. **R and/or Python** (Python ships with macOS): install R/RStudio
2. **An IDE or editor**
   - RStudio for R
   - VS Code for essentially any language, including R and Python
3. **An AI coding tool**, whichever you prefer: Claude Code, Codex, or Cursor

Optional: course materials live on GitHub, so skim an easy introduction to Git/GitHub from my MKT 615 course.

> If this is the **first time** you have heard any of these terms, raise your hand and ask questions.

Use of AI (LLMs)

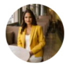
I **expect** you to use AI in this class. Learning to use it is an emerging skill. Keep the following in mind:

- AI is permitted to help you **brainstorm topics** or **revise work you have already written**
- Minimum-effort prompts produce low-quality results. Refining prompts to get good outcomes takes work
- Proceed with caution: **do not assume the output is accurate or trustworthy**

> **Hallucinations and mistakes are very common.**

For transparency, every student submits a **log of the LLM prompts** used on each assignment.

Why this course matters



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\+ Follow

AI PMs is the new hot job role. Netflix is paying $900K for AI PMs. Meta is paying $1M+.

Here are 8 skills every AI PM should have and a short course to start each one.

1️⃣ AI & ML Fundamentals
You don't need to train models, but you must understand how they work, their lifecycle, and their limits (hallucinations, bias, cost).
📚 Course: AI Product Management Specialization – Duke University (Coursera)

2️⃣ Applied AI Patterns & Tooling
Know agentic patterns (ReAct, Plan-and-Execute, CodeAct) and AI stacks (LLM APIs, vector DBs, LangChain, CrewAI).
📚 Course: IBM AI Product Manager Professional Certificate

3️⃣ Data Literacy & Evaluation
Be able to query, inspect logs, run A/B tests, and measure model quality (precision, recall, cost per token).
📚 Course: Generative AI for Product Managers Specialization

4️⃣ AI UX & Product Thinking
Design flows that balance creativity and control. Confidence scores, prompt chaining, guardrails.
📚 Course: AI for Product Management Course (Pendo)

5️⃣ Cross-Functional & Documentation Skills
Translate between engineers, data scientists, and business teams. Write PRDs with clear inputs, outputs, guardrails, and metrics.
📚 Course: Artificial Intelligence Product Certification (Product School)

6️⃣ AI Strategy & Business Acumen
Spot high-ROI AI use cases, evaluate competitors, and model trade-offs (latency, accuracy, cost).
📚 Course: Mastering Generative AI for Product Innovation (Stanford Online)

7️⃣ Experimentation & Deployment Mindset
Own offline + online evaluations. Know when to tweak UX, retrain models, or change data inputs.
📚 Course: AI Product Management Bootcamp (Maven)

8️⃣ Emerging Trends Awareness
Stay ahead on new models (Gemini, Claude, Mistral), multimodal AI, and evolving standards like MCP.
📚 Course: Elements of AI – University of Helsinki

Source

Grading

Component	Weight
4 individual assignments	60%
Semester-long project (groups of 5–6)	30%
Participation	10%
Total	**100%**

No final exam. Your grade is the weighted average above.

Assignments

- **4 individual assignments, 60%** of the grade
- Roughly **three weeks** for each

Each submission is an **R (Markdown/Quarto) or Python notebook**, converted to HTML or PDF, containing:

- Code, properly commented
- Outputs (figures, tables)
- Your answers to the assignment questions
- The **log of LLM prompts** used

> **Why a notebook?** Reproducibility. Someone else, including future you, has to be able to re-run it and get the same output and results.

The Semester Project

What it is

Analyze a **real-world dataset** to answer one or more marketing questions. For example:

- **Social media sentiment analysis**: how brand perceptions changed over time, which content performs better
- **Customer segmentation**: who are our most valuable segments?
- **Churn prediction and retention**: which users are likely to churn next month?

An example from last year

Exploring Customer Satisfaction Factors for Olist E-Commerce

Driving Pet Product Success

MKT 566
Group 8
Final Presentation

GROUP 5

CUSTOMER SHOPPING MARKETING ANALYSIS

SPOTIFY PROJECT

What Music To Produce?

WHAT DRIVES PROPERTY PRICES ACROSS U.S. MARKETS?

Data-Driven Insights for Real Estate Marketing

Examples from a colleague's class

ChatGPT and online content

Using Stack Overflow and YouTube data, students found a decline in Stack Overflow activity for coding topics (Python, Java) alongside an increase in AI-related videos on tech channels.

Read it

Examples from a colleague's class

Sentiment analysis of Sephora reviews

Using customer review data, students asked whether reviews help identify which Sephora products get featured as *trending* on the website.

Read it

Examples from a colleague's class

Breaking box office

Using Kaggle movie reviews and Box Office Mojo revenues, students documented a **null result**: customer reviews generally do not explain box office performance, though they may matter for less popular movies.

Read it

Examples from a colleague's class

Gender effects in a dating app

Using Bumble review data, students found that although more users appear to be male, women rate the app higher than men. A nice application of the `gender-guesser` package.

Where to find data

- Kaggle datasets, e.g. customer churn, shopping trends, Steam
- Yelp open dataset
- Amazon, Google reviews, Steam and more
- Marketing Science papers often ship data and code for replication
- **Collect it yourself** via an API

> I have data on **Airbnb, TripAdvisor, Expedia, and Yelp**. Ask me if you want it.

Deliverables and deadlines

Deliverable	Due
Form groups	September 15
Mid-term proposal slides (~15 min incl. Q&A)	October 12 (presented Oct 13, 15)
Notebook with cleaning and analysis (HTML or PDF)	December 3
Final presentation slides (~15 min incl. Q&A)	November 30 (presented Dec 1, 3)
Peer evaluations	December 15

Do you want me to form the groups, or would you rather do it yourselves?

Read the syllabus

Everything I have just discussed is in the **syllabus**. Please **read it at least once**.

Marketing Analytics

What it is

Marketing analytics is the practice of measuring, managing, and analyzing data from marketing activities to optimize business performance.

It covers:

1. Data collection and integration
2. Visualization and reporting
3. Advanced analysis
4. Defining, measuring, and tracking performance (KPIs)
5. Translating insights into decisions

1. Data collection and integration

- **Gathering** information from the web, social media, e-commerce platforms, and elsewhere
- **Unifying** datasets from different sources (online clicks, offline transactions) so they can answer a marketing question together

2. Visualization and reporting

Creating figures, dashboards, and reports that present insights to stakeholders:

- Advertising effect on sales
- Eco-labeling program growth
- Relationships between variables

3. Advanced analysis

Attribution modeling (causality)

Which channels or campaigns deserve credit for conversions?

Recommendation and segmentation (clustering)

Group customers by behavior or demographics to tailor messaging; identify products consumers are likely to buy; product placement

Predictive modeling (regression, machine learning)

Forecast campaign performance; predict churn; identify fraudulent activity such as fake reviews, accounts, and clicks; forecast prices

4. KPIs

Metric family	Examples
Acquisition	Cost per click, cost per acquisition
Engagement	Click-through rate, time on site
Conversion	Lead-to-customer rate, average order value
Retention	Churn rate, customer lifetime value
Incrementality	How much of the sales lift is *attributable* to the campaign

5. Translating insights into decisions

Turning analysis into action:

- **Reallocating** budget to top-performing channels
- **Optimizing** messaging and creative
- **Changing** the landing page
- **Identifying** and leveraging new trends
- **Optimizing** prices

In a nutshell

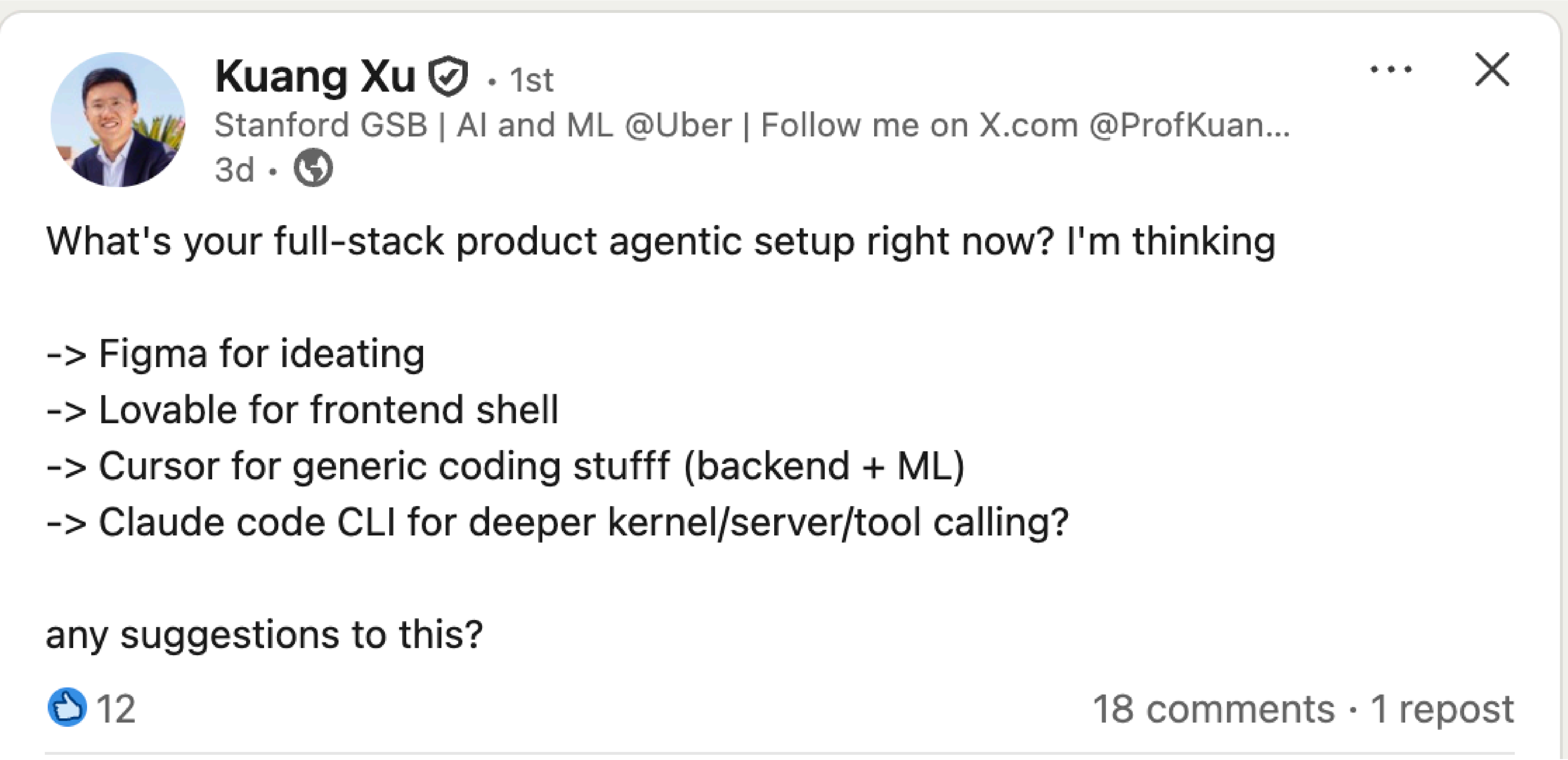
> By systematically applying statistical methods and data-driven storytelling, marketing analytics enables organizations to **understand what works**, **identify growth opportunities**, and **maximize their return on marketing investment**.

The Course Content

What we will cover

- Data analysis and visualization
- Regressions
- Clustering and recommendations
- Classifiers
- Causality: experiments and observational data
- **LLMs and agents**: new challenges for brands (e.g. search), and how firms are leveraging them

What we will cover



Kuang Xu • 1st

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What's your full-stack product agentic setup right now? I'm thinking

-> Figma for ideating

-> Lovable for frontend shell

-> Cursor for generic coding stufff (backend + ML)

-> Claude code CLI for deeper kernel/server/tool calling?

any suggestions to this?

12

18 comments · 1 repost

Source

What we will *not* cover

- **This is not a programming course.** Learning to code is mostly up to you and your LLM
- **This is not an advanced statistics, econometrics, or ML methods course.** The emphasis is on **data-driven decision-making**, so we stay at a high level conceptually

> I assume you are familiar with **basic statistics and probability**. I can post review slides if that would help.

Questions?